

Advancing 3D CAD with Workflow Graph-Driven Bayesian Command Inferences

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ABSTRACT

Advancements in 3D generative AI have significantly improved design capabilities, particularly for creating 3D objects and environments. However, the focus of Generative AI on mesh-based models limits opportunities for detailed modifications. Accurate and complex 3D modeling is crucial for manufacturing, which requires high precision and considerable mental effort. This complexity often leads to variability in efficiency among designers, with some employing faster and more accurate techniques and others using less efficient workflows. This variation undergoes the need to optimize modeling sequences. By inferring a user's intended designs, tailored commands and sequences can be suggested to enhance the precision of 3D modeling. Addressing this, we propose a system that predicts user modeling steps using an inference model based on behavior, thereby promoting efficient workflow and precise command usage. User studies demonstrate that our system minimizes modeling errors, streamlines processes, and offers recommendations for effective command usage.

CCS CONCEPTS

• **Applied computing** → **Computer-aided design**; • **Human-centered computing** → **Interactive systems and tools**; **Command line interfaces**.

KEYWORDS

Computer-Aided Design, 3D Modeling Workflow, Computational Design, Design Command Inference, Bayesian Information Gain

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1 INTRODUCTION

Recently, interest in 3D generative artificial intelligence (AI) has surged, leading to an increase in research utilizing this technology in the field of design [1, 5, 10]. Advancements in this technology

have opened up new possibilities for designers, especially in assisting the design process for three-dimensional (3D) objects and environments. Design professionals have recognized the potential of integrating text-to-image AI tools such as 3DALL-E to generate reference images, renderings, materials, and design inspiration[10]. Furthermore, tools such as SPAGHETTI[4] and SALAD[7] augment the design process by enabling the rapid generation, synthesis, and exploration of 3D mesh models by mixing and refining 3D shapes. However, since generative AI mainly produces mesh-based 3D models, detailed design modifications are limited. Thus, the use of Generative AI to transform mesh-based 3D models is highly valued, particularly for providing inspiration in the early stages of design or creating digital-only assets[10]. Precision control over surfaces is imperative in real-world 3D model creation, particularly for manufacturing purposes. Accurate modeling is vital because it affects the precision and comprehension of the model, and is actively used in maintaining large-scale industrial facilities[9]. Such modeling, which requires complex and precise outcomes, can be mentally demanding and time consuming. Additionally, different individuals may adopt various approaches to create the same 3D model. While some people may utilize efficient commands and procedures for faster and more accurate modeling, others may engage in less efficient processes. This suggests the presence of an optimal modeling sequence, offering a more efficient alternative to those using inefficient methods. If we can determine what a user wants to create by analyzing data from multiple users, we can recommend an efficient modeling sequence and commands for precise 3D modeling, thus reducing the cognitive load on users. Therefore, we propose a system that identifies design modeling steps based on user behavior, offering an efficient modeling workflow and commands. To support conventional design processes for designers, we conducted research based on Rhinoceros7 (hereafter Rhino7), which supports NURBS-based modeling. This addresses the limitations of mesh-based models and is actively used in the 3D modeling industry for architecture and product design. This study involves the following tasks: 1) developing a Rhino plugin to record user-entered commands and collect 3D CAD modeling sequence data.; 2) application of a W-graph with added command nodes for multiple fixed modeling tasks ; 3) measurement of similarity between 3D models for clustering of nodes in the W-graph; 4) Utilizing the BIG(Bayesian Information Gain) framework to infer 3D modeling processes; and 5) experimenting with a modeling sequence recommendation system.

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2 RELATED WORKS

2.1 Design aid for 3D modeling

Extensive studies have focused on enhancing 3D model design [1, 4, 6, 7, 13]. Marzal et al. (2020) presented a Computer-Based Generative Design (CGM) [1] that enables designers to efficiently visualize and transform sketches and ideas. CGM fosters rapid exploration of diverse design options and innovative concepts. Furthermore, CGM [1] expands the design range by facilitating the manipulation and modification of complex forms and structures, thereby empowering designers to develop more creative and innovative solutions. However, the complex algorithms and computational intensity of CGM require significant investments in time and effort for users to master and exploit their full potential. As this technology is based on Grasshopper, it occasionally leads to excessive computational loads, which can adversely affect the efficiency of the design process.

Advancement in 3D generative AI enables the generation of part-level 3D models. For instance, Spaghetti [4] and SALAD [7] enrich design diversity by synthesizing and interpolating distinct segments of 3D objects at the part level. SPAGHETTI [4] employs a Decomposition Network that breaks down the shape embedding z into distinct 3D part embeddings aligned with a Gaussian mixture model (GMM), obtains mixed embeddings through a mixing network, and finally determines the 3D form in the Occupancy Network. The Decomposition Network enables part-level control by allowing localized transformation, amalgamation, and interpolation of segments from diverse shapes. SALAD [7] enhances the methodology established by SPAGHETTI [4], thereby yielding more sophisticated 3D shapes. It integrates a text-guided part-completion mechanism and affords advanced editing functionalities for the precise mixing and modification of parts. Although these investigations provide an array of design perspectives utilizing 3D generative AI and bolster the preliminary stages of the design process, they have limitations in supporting the mid to late stages of actual production and concrete design development. Therefore, methodologies that augment the efficacy of conventional CAD for meticulous 3D model construction in intermediate-to-advanced design phases are required.

Schulz et al. [13] introduced a system utilizing parametric CAD data to explore and optimize design shapes in real time, aiding experts to efficiently search the design space and lower design barriers for general users. This system capitalizes on pre-calculated data and is a new interpolation method for rapid parameter adjustments and model optimization in real time. Furthermore, Chang et al. [2] proposed the concept of a W-graph, which encodes the similarities and differences in the approaches taken by users in various modeling demos of a fixed task. The primary aim is to support the 3D modeling learning process for novice designers, while also proposing unconventional modeling methods for experienced users. The W-graph, which is composed of nodes with metadata (command log data, screen recording, scene graph snapshot) and edges with event data (including timestamped command invocations). This structure allows understanding of multiple users' modeling processes by observing equivalent-intermediate nodes. Each directional edge from the start node to the end node represents a potential workflow for completing the task, and multiple edges

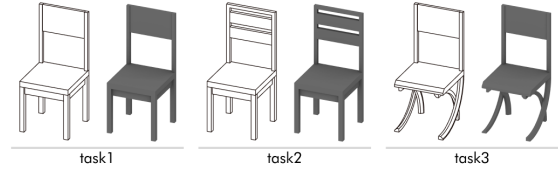


Figure 1: 3D chair model of the designed tasks for the preliminary experiment

between two states represent alternative approaches to the task segmentation. The W-graph serves as a backend for applications suggesting alternative workflows by expressing the workflow structure. Accordingly, they presented potential applications such as W-Suggest, W-Guide, and W-Instruct systems that can be implemented through a W-graph, demonstrating their potential to support user design processes. In our research, we employed the W-graph concept to encode the workflows of various users engaged in distinct fixed tasks, integrating it as the backend of the system.

2.2 Bayesian Information Gain Framework

The Bayesian Information Gain (BIG) framework is designed to model the information gain between user input Y , system feedback X , and the user's intended goal Θ . Its primary objective is to decode user intent and optimize information acquisition in user-computer interactions. The system predicts user responses based on prior knowledge $P(\Theta)$, and a user behavior model $P(Y|\Theta, X)$, updating its understanding and calculating information gain accordingly. This framework enables the system to adapt its feedback to maximize expected information gain, thereby enhancing user interaction efficiency. Essentially, the BIG framework serves as a methodology for deducing the focus of user interest based on user behavior. Consequently, it has been widely adopted in various studies to facilitate search processes [8, 11, 12, 14, 15]. For example, BIGnav [11] interprets user input to update the view with the maximum expected information gain, aiding in user navigation.

Liu et al. employed the BIG framework in the development of BIGfile [12], effectively streamlining file searches by providing shortcuts that reduce the number of steps (user inputs) required to locate a specific file or folder. Upon receiving user input, the BIGfile algorithm updates $P(\Theta)$ using Bayes' rule and identifies the target with the highest possible information gain from all potential target nodes N combinations. This maximizes the expected information gain from the user's next input and optimizes the file search path. Specifically designed for hierarchical tree structure, the BIGFile system enhances the user's navigation by recommending file paths that maximize the expected information gain, thus making navigation more efficient and faster. In our study, we aim to determine the modeling stage that maximizes the expected information gain from the user's next input (specifically, the commands to be applied on a given edge) based on the intricate tree structure of the W-graph. Through this approach, we seek to predict the most likely modeling outcome desired by user and offer them targeted command recommendations to efficiently achieve the modeling stage.

3 PROPOSED METHOD

3.1 Overview

The proposed system aims to infer user intent and identify the most efficient workflow for the current modeling stage, then offer a suitable command sequence. The system undergoes the following stages. **1) W-Graph Generation:** This initial phase extracts modeling sequence data from multiple users completing three fixed tasks. A W-graph was constructed, consisting of nodes representing 3D model snapshots and edges signifying the recorded commands. This W-graph was encoded by utilizing the commands and snapshot data derived from each modeling process. **2) 3D Model Similarity Calculation:** To merge identical nodes of a single user and create equivalent intermediate nodes observed in the modeling sequences of multiple users, a part-level similarity analysis is conducted. **3) Inference Method:** Using the W-graph, the system evaluates the modeling stage offering the highest expected information gain from the current position. This facilitates the modeling process by recommending a step predicted to lead to the intended stage. **4) Interfaces:** The modeling sequence from the current stage to the suggested stage is visualized through a W-graph and provided as a plugin for Rhino 7.

3.2 W-graph Generation

To observe the various user approaches to fixed tasks, three specific tasks were designated, as depicted in Figure1. To analyze the convergence and divergence in the modeling sequence, we selected three types of chairs that included elements of a partially common design. For the experiment, design students who frequently use 3D modeling tools were recruited for the experiment. Fifteen participants (eight male, seven female, ages 20-24, average Rhino usage period=21.2 months) took part, with five per task, to carry out the modeling. A plugin was developed for Rhino7, a NURBS-based modeling tool, to record the user modeling process. Each time a user entered a command, the command, time, and snapshots were recorded. These snapshots were then used to encode the W-graph, following the methodology outlined in 'Workflow to Graph Construction'[2].

Step 1. Preprocessing Each user's event sequence is represented as a set of nodes (one node per event), with directional edges connecting each event in timestamp order (Figure2-a). If a snapshot 3d file conversion does not generate a mesh (e.g., if only curves exist or nothing at all), no node is created, and command information is added to the previous edge. Each demonstration begins with a blank document and ends with a completed 3D model; therefore, a START node with directed edges is added to the first node of each demo, and the last node of each demo is merged into an END node (task1, task2, task3) (Figure2-b). Each demonstration depicts a unique directional path from a START to an END node.

Step 2. Collapsing Node Sequences Within each path, nodes with similarity values below a threshold value are merged (Figure2-c), as detailed in '3.3.3D Model Similarity Calculation'. In addition, the command data of the edges connected to the merging nodes were added to the incoming edge of the previous node being merged.

Step3. Sequence Merging This step involves creating "equivalent intermediate" nodes, representing intermediate stages in the task, used to depict common intermediate states appearing in different

users' work (Figure2-d). To achieve this, the similarity between nodes of different users is compared, employing the Model Similarity Calculation method.

Step4. Add command nodes Each edge was replaced with a W-graph composed of command nodes, based on the command information contained within each edge.(Figure2-e). At the end of these steps, we can encode the modeling sequences of multiple users for Tasks 1, 2, and 3, as obtained through the experiment, into a W-graph, as shown in Figure3.

3.3 3D Model Similarity Calculation

In the W-graph generation phase, the decomposition network from Spaghetti [4]] is utilized to analyze the similarity between snapshots by decomposing the 3D model of each node. This decomposition breaks down shape embedding z into 16 part-level representations z_{id} . This method decomposes a snapshot node's 3D model into 16 Gaussian blobs, as shown in Figure4-a. Each part of z_{id} is represented by extrinsic geometric information (g_i), indicating the location of each Gaussian blob, and intrinsic geometric information (s_i), denoting the shape characteristics. The set of extrinsic parameters is represented as $e_i = \{c_i, \lambda_{i1}, \lambda_{i2}, \lambda_{i3}, u_{i1}, u_{i2}, u_{i3}, i\}$, including the blending weight $i \in \mathbb{R}^1$, the mean of the GMM $c_i \in \mathbb{R}^3$, eigenvalues and eigenvectors of the covariance matrix $u_{ij} \in \mathbb{R}^3$, $\lambda_{ij} \in \mathbb{R}^1$. Similar intracategory parts of the dataset were consistently represented using the same Gaussian blobs, facilitating similarity calculations. This means that identical 3D models could be represented using the same Gaussian blobs. To calculate the part-level similarity, we used the c_i coordinates from g_i , represent the center location of each Gaussian blob. Specifically, we calculated Euclidean distances by matching the ordered 16 Gaussian blobs to those of another node and averaging the distances. Nodes with distances below a specified threshold were considered to represent the same 3D model(Figure4-b).

3.4 Inferencing Algorithm

This Inference Algorithm is based on the BIG algorithm, which employs the BIGFile algorithm and a hierarchical tree structure. Θ represents all possible target nodes, and the aim of the system is to identify the node θ that corresponds to the modeling stage the user wishes to create. The probability $P(\Theta)$ for all potential target nodes Θ indicates the likelihood of each node being the intended target. This probability is continuously updated based on user behavior and interaction history. After each user input, $P(\Theta)$ is updated using Bayes' rule. X represents the complete set of all modeling sequences that lead to the modeling stage recommended to the user, where x signifies the modeling sequence pertaining to an individual node. This set includes the current node S and the intermediate stage S , identified as having the highest expected information gain in reaching the N recommended modeling stages, as determined by the BIGFile algorithm. At every stage, the user provides input $Y = y$, which signifies the action of executing a command to progress to a subsequent node. According to [3], users exhibit an accuracy rate of nearly 94% in their interactions. Conversely, in approximately 6% of instances, they may select an incorrect option, such as clicking on the wrong folder. As a result, the probability $P(Y = y|\Theta, X = x)$ represents the likelihood of a user correctly reporting input $Y = y$

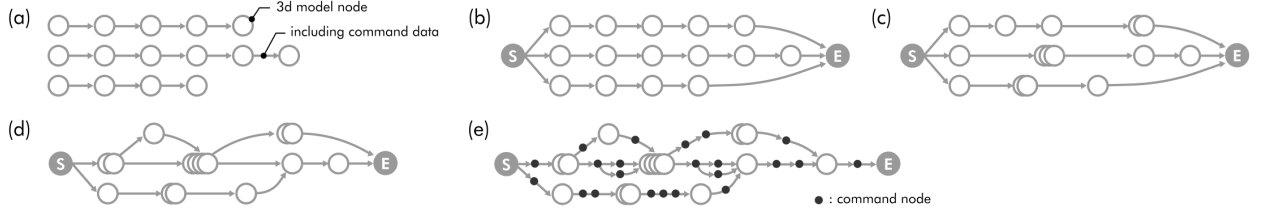


Figure 2: Illustration of how sequences get compressed and merged into a W-graph

when presented with $X = x$ is set at 0.94 for correct inputs and 0.06 for incorrect ones. Here, S denotes the node with the highest similarity when the user successfully executes a command. This is calculated using the 3D Model Similarity Calculation method and employs the same threshold value ϵ used in the construction of the w-graph. After each user inputs, S is updated to reflect these changes. Subsequently, the algorithm then recommends the intermediate nodes that offer the highest expected information gain in relation to the N probable nodes identified by the BIGFile algorithm, thereby facilitating efficient user navigation towards the target node.

Algorithm 1 Inferencing Algorithm

Require: $\Theta, X, Y, P(Y = y|\Theta, X = x), IG_{\max} = 0$

Ensure: Return set A that, together with set S , has the maximal expected information gain (IG).

- 1: Receive user input $Y = y$
- 2: Find most similar node (within w-graph) S
- 3: Update the probability distribution of Θ (Bayes rule):

$$P(\Theta|X = x, Y = y) = \frac{P(Y = y|\Theta, X = x)P(\Theta)}{P(Y = y|X = x)}$$

- 4: **for** All combinations of N nodes under the hierarchy of the current node **do**
 - 5: Compute $IG(S \cup A) = I(\Theta; Y|X(S \cup A))$
 - 6: $= H(\Theta) - H(\Theta|X(S \cup A), Y)$
 - 7: **if** $IG(S \cup A) > IG_{\max}$ **then**
 - 8: $IG_{\max} = IG(S \cup A)$
 - 9: $A_{\max} = A$
 - 10: **end if**
 - 11: **end for**
 - 12: **return** $A_{\max}$
-

3.5 Interface

Our system identifies the node corresponding to the user's current 3D model state through a similarity comparison process. Based on the user's actions (command inputs), it then provides an appropriate modeling sequence and commands to reach the recommended modeling stage, thereby supporting the modeling process. The interface enables users to identify popular modeling stages that are frequently used, as reflected in the equivalent intermediate nodes on the W-graph. Furthermore, by examining multiple edges connected to their current state, users can investigate alternative modeling pathways (Figure 5-a).

4 IMPLEMENTATION AND RESULTS

4.1 Experimental Design

To assess the effectiveness of the recommendation algorithm based on the generated workflow, we enlisted four participants skilled in Rhino. The experiment was structured as follows: an introduction, consent, and tutorial session (lasting approximately 10 min), followed by three separate modeling tasks (Modeling 1, Modeling 2, and Modeling 3 (each task took approximately 15 min)) and concluding with an in-depth interview (20 min). The tasks corresponded to Tasks 1, 2, and 3 (as specified in 1) from the previous experiment. Throughout each session, the participants' command inputs were recorded, and the recommendation interface was updated following each successful command execution. Following the completion of all experiments, we conducted interviews to gather feedback and assess the participants' overall experience with the modeling process using this algorithm.

4.2 Results and Discussions

After completing the three designated modeling tasks, we conducted interviews with the participants to evaluate the effectiveness of our inference system by comparing its strengths and weaknesses with those of Rhino7 as the baseline. All participants unanimously agreed that the system reduced errors and improved the efficiency of the modeling process. For instance, P4 observed that the system efficiently minimized repetitive actions commonly performed by designers. A habitual sequence of drawing a 'polyline' and then using 'extrudecrv,' for example, could be streamlined by using the more efficient 'box' command. Additionally, three out of the four participants mentioned that they found the system more helpful when executing more complex models, particularly in Task 3 (Figure 5-b). In fact, the time taken for modeling compared with the data collection phase, with a 26.5% greater reduction, was observed for Task 3 compared with the baseline. P1 emphasized the system's consistent recommendation of often-forgotten commands, such as 'cap,' which proved crucial in preventing overlooked steps to convert 3D objects into closed surfaces, thereby reducing the need for subsequent modifications. P3 appreciated that the W-graph offered not only provided the desired outcomes but also potential alternative design directions that could potentially influence the final design. However, in two experiments, there were instances in which recommendations could not be provided owing to insufficient data. All participants suggested that adding explanations for commands could assist beginners in learning, while visualizing

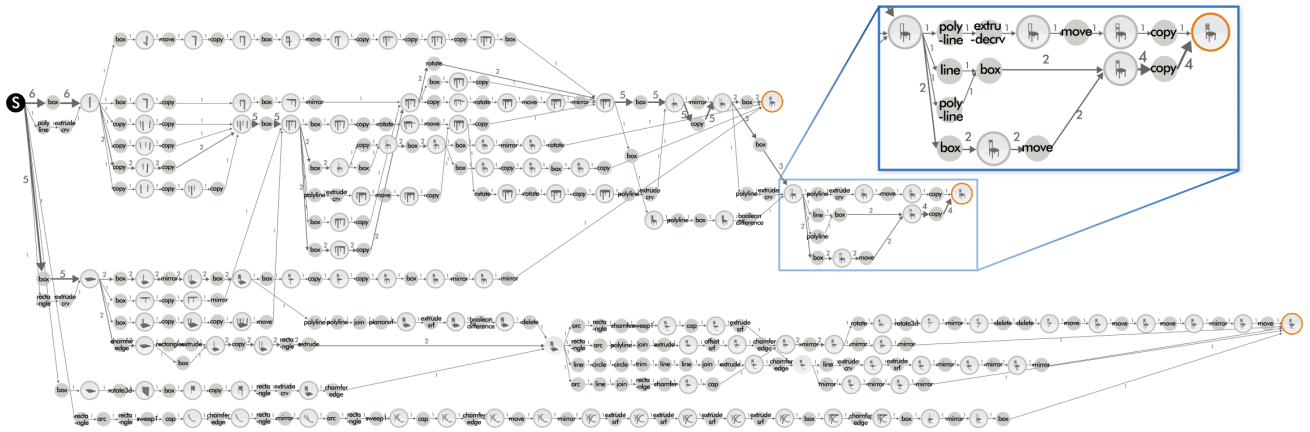


Figure 3: W-graph encoding the modeling process of Tasks 1, 2, and 3

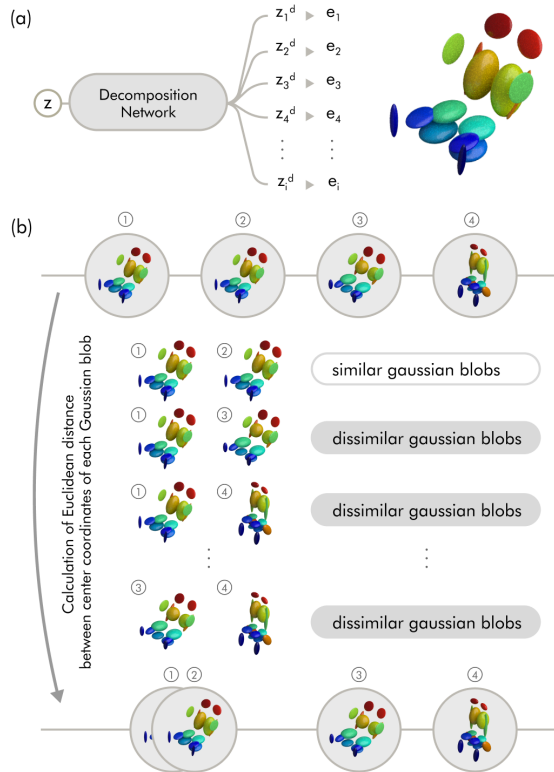


Figure 4: 3D model similarity calculation process

each step with highlighted changes could enhance their understanding of the modeling process. P4 emphasized that the system not only helped in understanding the appropriate use of commands but also the modeling sequence itself. Regarding the visualization of recommendations, they noted that showing the modeling sequence from the current stage to a distant stage aided in understanding the

overall process. Particularly, when the goal is clear, following the sequence can reduce mental load. However, concerns were raised that visualizing stages too far ahead might cause confusion, as most users only reference up to two to three steps later. In conclusion, the implementation of this recommendation system showcased potential for increased efficiency by saving modeling time and enhancing accuracy, effectively assisting designers in handling complex tasks. However, there are limitations owing to data insufficiency in presenting modeling directions using actual modeling data. Additionally, there is a notable preference among users for simpler and more visually intuitive graphs, underscoring the need for improvements in the clarity and visual appeal of these graphs.

5 CONCLUSION

This research streamlines the modeling process for designers by predicting modeling goals based on user behavior and recommending steps to achieve these goals. By viewing the W-graph of a specific part, users can comprehend the modeling sequence and identify the necessary commands for actual production. This approach helps reduce errors and promotes efficiency in modeling throughout the process. Participants in the experiment reported that the system reduced their cognitive load during modeling, decreased modeling time, and improved accuracy, especially in complex modeling tasks. Additionally, the system demonstrated the potential for educational use, aiding novices in learning the 3D modeling process. Users can effectively develop 3D modeling skills by observing recommended commands, modeling progression, and alternative approaches. Designed to support user-modeling processes for Rhino7, this system demonstrates versatility owing to its ability to be applied to other 3D modeling environments, provided that the modeling sequence data for drawing a W-graph is secured. This adaptability enhances its utility as a support system for various 3D modeling platforms. However, concerns have been raised regarding the potential confusion caused by displaying nodes from stages too far ahead in the W-graph interface. Therefore, to provide precise information and support to users, the interface should be modified to display outcomes from the current step up to only the next step, or up to

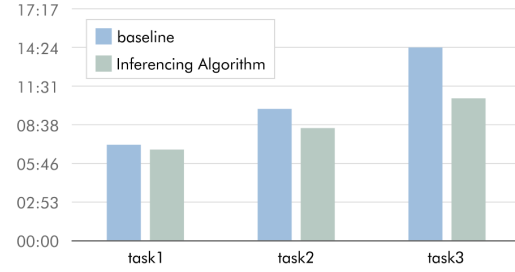
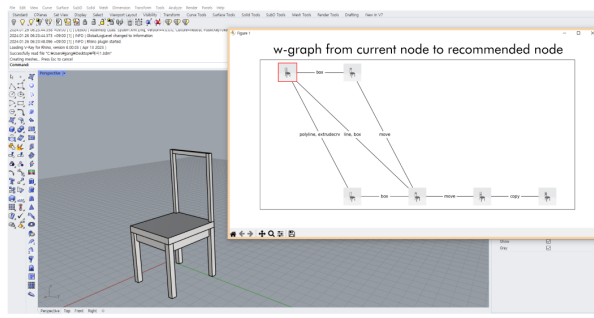


Figure 5: Proposed system interface (left), and task completion time (right)

two to three steps later. This adjustment would ensure more accurate knowledge transfer and more effective modeling assistance for users.

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REFERENCES

- [1] Jorge Alcaide-Marzal, Jose Antonio Diego-Mas, and Gonzalo Acosta-Zazueta. 2020. A 3D shape generative method for aesthetic product design. *Design studies* 66 (2020), 144–176. <https://doi.org/10.1016/j.destud.2019.11.003>
- [2] Minsuk Chang, Ben Lafreniere, Juho Kim, George Fitzmaurice, and Tovi Grossman. 2020. Workflow graphs: A computational model of collective task strategies for 3D design software. In *Graphics Interface 2020*. <https://openreview.net/forum?id=qXEzq5agzIN>
- [3] Stephen Fitchett and Andy Cockburn. 2015. An empirical characterisation of file retrieval. *International Journal of Human-Computer Studies* 74 (2015), 1–13. <https://doi.org/10.1016/j.ijhcs.2014.10.002>
- [4] Amir Hertz, Or Perel, Raja Giryes, Olga Sorkine-Hornung, and Daniel Cohen-Or. 2022. Spaghetti: Editing implicit shapes through part aware generation. *ACM Transactions on Graphics (TOG)* 41, 4 (2022), 1–20. <https://doi.org/10.1145/3528223.3530084>
- [5] Ka-Hei Hui, Ruihui Li, Jingyu Hu, and Chi-Wing Fu. 2022. Neural wavelet-domain diffusion for 3d shape generation. In *SIGGRAPH Asia 2022 Conference Papers*. 1–9. <https://doi.org/10.1145/3550469.3555394>
- [6] R Kenny Jones, Paul Guerrero, Niloy J Mitra, and Daniel Ritchie. 2023. ShapeCoder: Discovering Abstractions for Visual Programs from Unstructured Primitives. *arXiv preprint arXiv:2305.05661* (2023). <https://doi.org/10.48550/arXiv.2305.05661>
- [7] Juil Koo, Seungwoo Yoo, Minh Hieu Nguyen, and Minhyuk Sung. 2023. Salad: Part-level latent diffusion for 3d shape generation and manipulation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 14441–14451. <https://doi.org/10.1109/icc51070.2023.01328>
- [8] Seung Won Lee, Hwan Kim, Taeha Yi, and Kyung Hoon Hyun. 2023. BIGaze: An eye-gaze action-guided Bayesian information gain framework for information exploration. *Advanced Engineering Informatics* 58 (2023), 102159. <https://doi.org/10.1016/j.aei.2023.102159>
- [9] Ke Li, Susanne Schmidt, Tim Rolff, Reinhard Bacher, Wim Leemans, and Frank Steinicke. 2023. Magic nerf lens: Interactive fusion of neural radiance fields for virtual facility inspection. *arXiv preprint arXiv:2307.09860* (2023). <https://doi.org/10.48550/arXiv.2307.09860>
- [10] Vivian Liu, Jo Vermeulen, George Fitzmaurice, and Justin Matejka. 2023. 3DALL-E: Integrating text-to-image AI in 3D design workflows. In *Proceedings of the 2023 ACM designing interactive systems conference*. 1955–1977. <https://doi.org/10.1145/3563657.3596098>
- [11] Wanyu Liu, Rafael Lucas d'Oliveira, Michel Beaudouin-Lafon, and Olivier Rioul. 2017. Bignav: Bayesian information gain for guiding multiscale navigation. In *Proceedings of the 2017 CHI conference on human factors in computing systems*. 5869–5880. <https://doi.org/10.1145/3025453.3025524>
- [12] Wanyu Liu, Olivier Rioul, Joanna McGrenere, Wendy E Mackay, and Michel Beaudouin-Lafon. 2018. BIGFile: Bayesian information gain for fast file retrieval. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. 1–13. <https://doi.org/10.1145/3173574.3173959>
- [13] Adriana Schulz, Jie Xu, Bo Zhu, Changxi Zheng, Eitan Grinspun, and Wojciech Matusik. 2017. Interactive design space exploration and optimization for cad models. *ACM Transactions on Graphics (TOG)* 36, 4 (2017), 1–14. <https://doi.org/10.1145/3072959.3073688>
- [14] Kihoon Son, Kyungmin Kim, and Kyung Hoon Hyun. 2022. BIGexplore: Bayesian information gain framework for information exploration. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–16. <https://doi.org/10.1145/3491102.3517729>
- [15] Kihoon Son, Seung Won Lee, Wondeuk Yoon, and Kyung Hoon Hyun. 2022. CreativeSearch: Proactive design exploration system with Bayesian information gain and information entropy. *Automation in Construction* 142 (2022), 104502. <https://doi.org/10.1016/j.autcon.2022.104502>